Note 2.3 The Precise Definition of a Limit

Definition of Limit

- **Definition:**
  Let $f(x)$ be defined on an open interval about x_0 except possibly x_0 itself, then we say the limit of $f(x)$ as x approaches x_0 is the number L and write

$$\lim_{x \to x_0} f(x) = L$$

  if for any $\epsilon > 0$, we can find a $\delta > 0$ for all x, when

$$0 < |x - x_0| < \delta$$

  there will be

$$|f(x) - L| < \epsilon$$

- **Remarks:**
  - The smaller of ϵ, the closer of $f(x)$ to L, thus we can restrict ϵ to $0 < \epsilon < 1, 0 < \epsilon < 0.5, \cdots$ but not $\epsilon > 0.1, \epsilon > 0.01, \cdots$
  - Since ϵ is arbitrary, so is $\frac{1}{2}\epsilon, \epsilon^2, \cdots$, thus $|f(x) - L| < \epsilon$ can be replaced by $|f(x) - L| < \frac{1}{2}\epsilon, \epsilon^2, \cdots$
  - The value of δ depends on the value of ϵ, but δ is not a function of ϵ. Whenever $0 < |x - x_0| < \delta$, we have $|f(x) - L| < \epsilon$. Whenever $0 < |x - x_0| < 0.5\delta, 0.003\delta, \cdots$, we still have $|f(x) - L| < \epsilon$.

Finding Deltas Algebraically for Give Epsilons

- **Strategy:**
  - Solve $|f(x) - L| < \epsilon$ to find (a, b) containing x_0.
  - Finding a value of $\delta > 0$ that places $(x_0 - \delta, x_0 + \delta)$ inside (a, b).
- **Example:**
  The example of strategy

Properties of Limit

- If $\lim_{x \to c} f(x)$ exists, the limit is unique.
- If the limit $\lim_{x \to c} f(x)$ exists, then $f(x)$ is locally bounded. That is when $\lim_{x \to c} f(x) = A$, then there exists $M > 0$ and $\delta > 0$ such that when $0 < |x - c| < \delta$, $|f(x)| \leq M$.(where $M = \max\{|L + \epsilon|, |L - \epsilon|\}$)
- Locally sign-preserving property:

    - Definition: If the limit $\lim_{x\to c} f(x) = A > 0$, then there exists $\delta > 0$ such that when $0 < |x - c| < \delta$, $f(x) > 0 \left(\text{or } f(x) > \frac{A}{2} \text{ when } \epsilon = \frac{A}{2}\right)$. If the limit $\lim_{x\to c} f(x) = A < 0$, then there exists $\delta > 0$ such that when $0 < |x - c| < \delta$, $f(x) < 0 \left(\text{or } f(x) < \frac{A}{2} \text{ when } \epsilon = \frac{A}{2}\right)$. That is, on the neighborhood of c except possibly at c, $f(x)$ and A have the same sign.
    - Corollary: If on a neighborhood of c except possibly at c, $f(x) \geq 0$ and $\lim_{x\to c} f(x)$ exists, then $\lim_{x\to c} f(x) \geq 0$.(It can be proved by contradiction)
- The composite function $f(x)$ is defined on a neighborhood of x_0 except possibly at x_0, if $\lim_{x\to x_0} g(x) = u_0, \lim_{u\to u_0} f(u) = L$ and $g(x) \neq u_0$, then $\lim_{x\to x_0} f(g(x)) = L$.
    - Remarks: The condition $g(x) \neq u_0$ on the neighborhood is necessary. Otherwise, the above result is false.